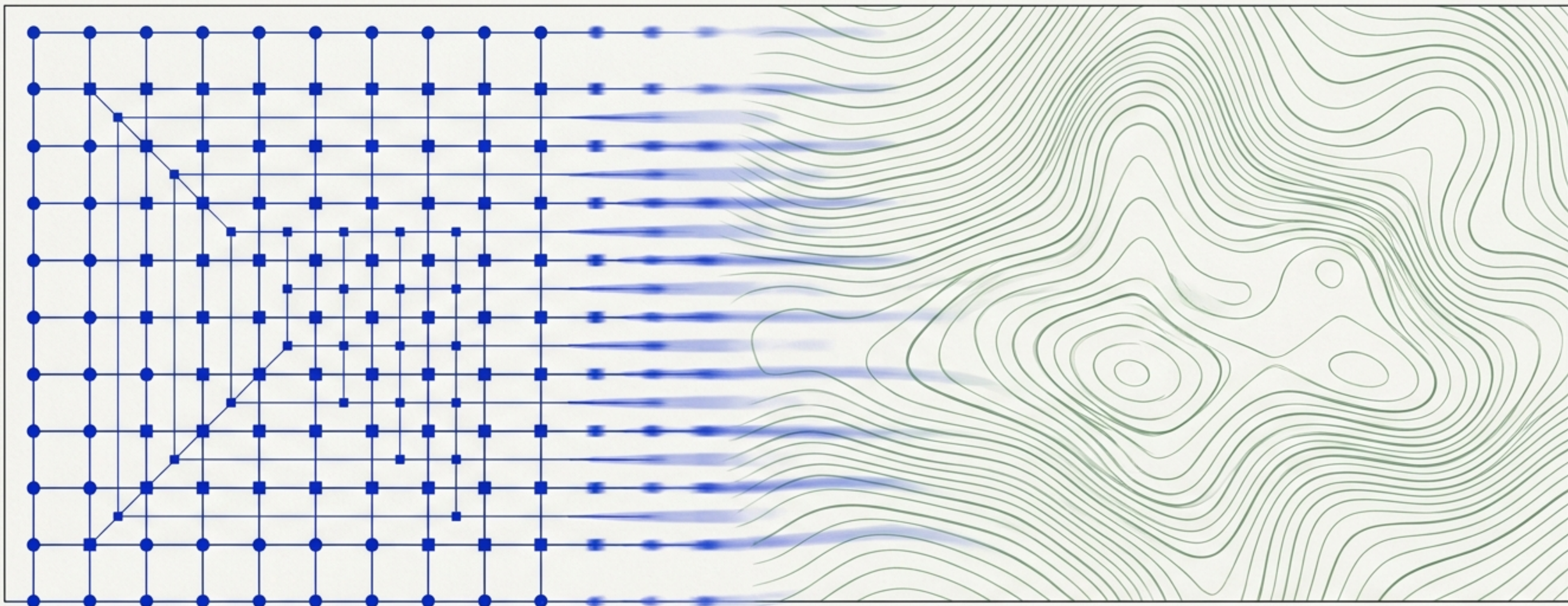


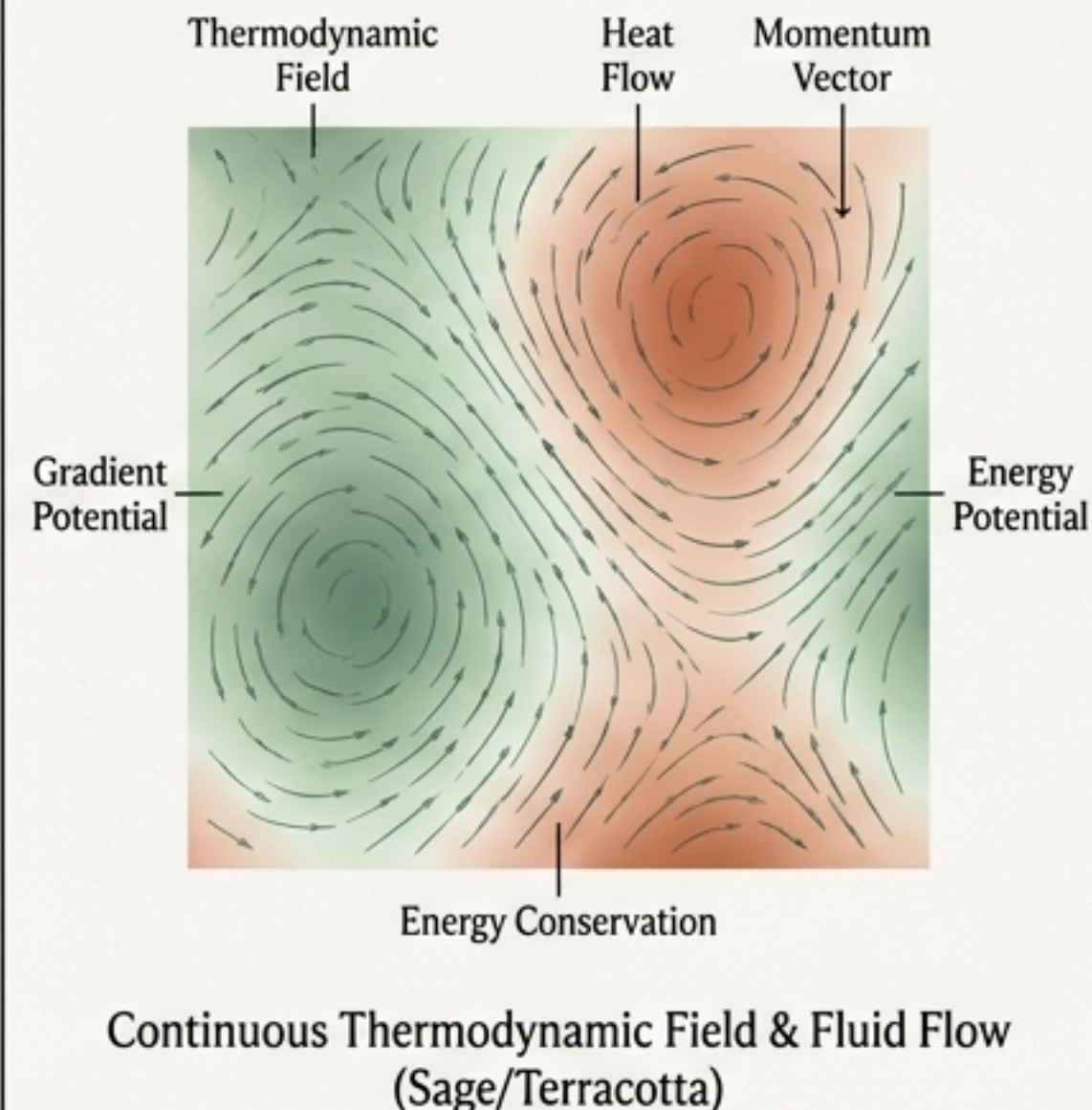
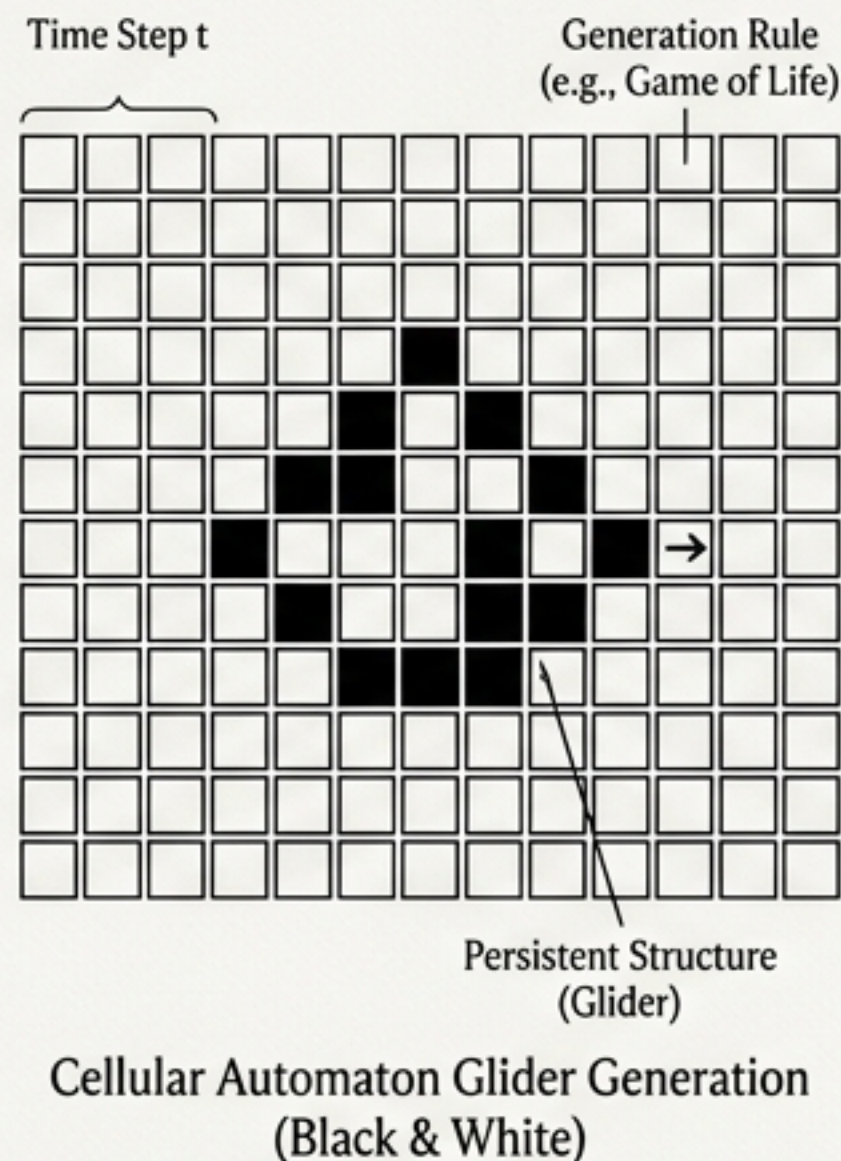
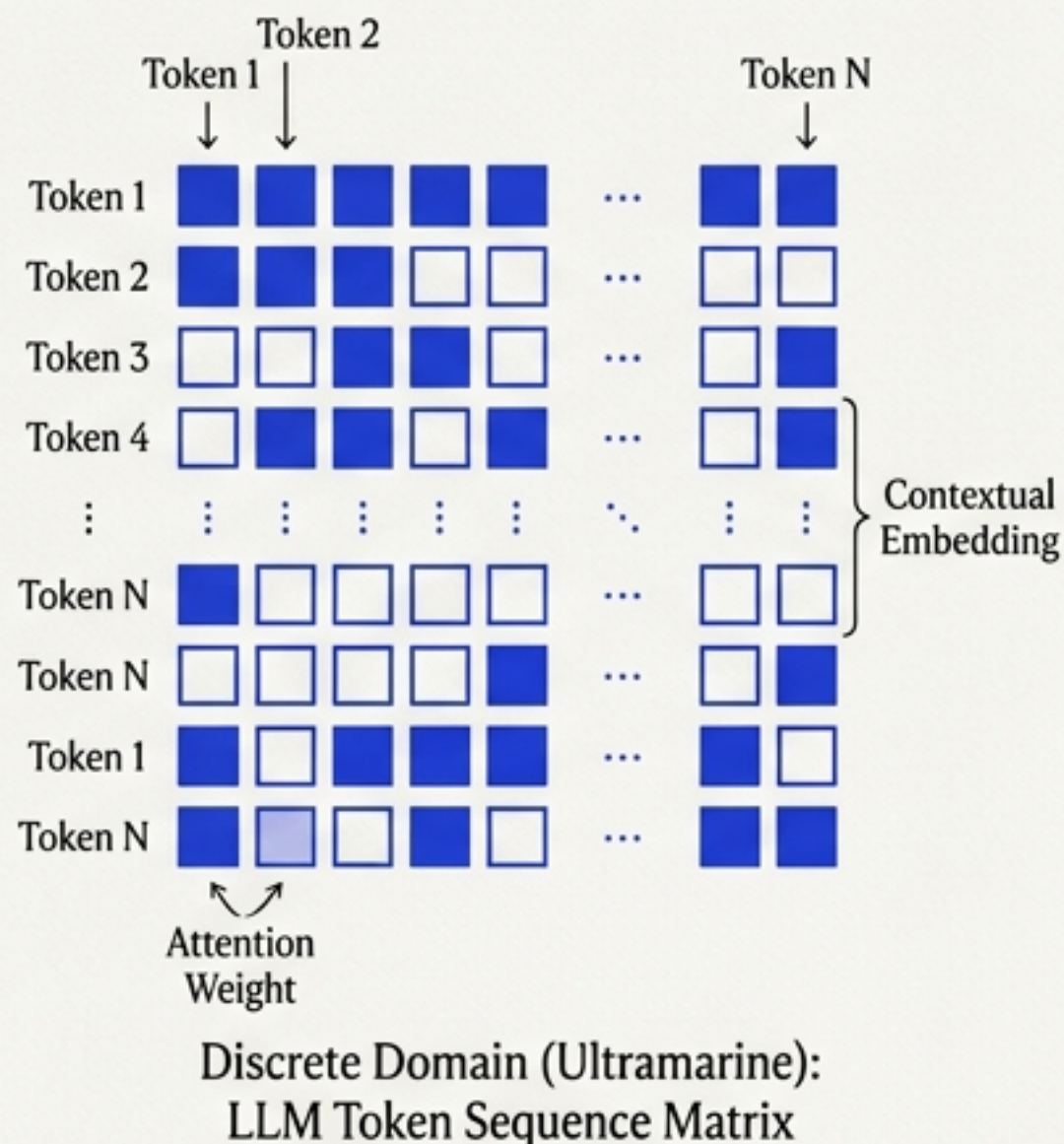


The Geometry of Distinction

A Unified Framework Linking Autoregressive Minds,
Continuous Field Dynamics, and Constraint-First Ontology.

Based on the theoretical
framework by Flyxion
(August 2025).

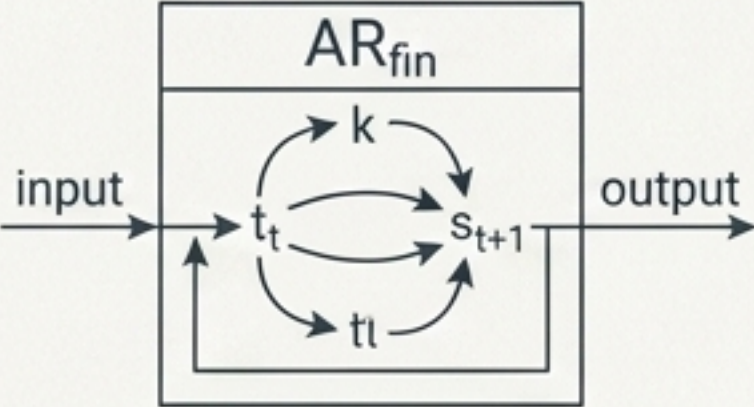
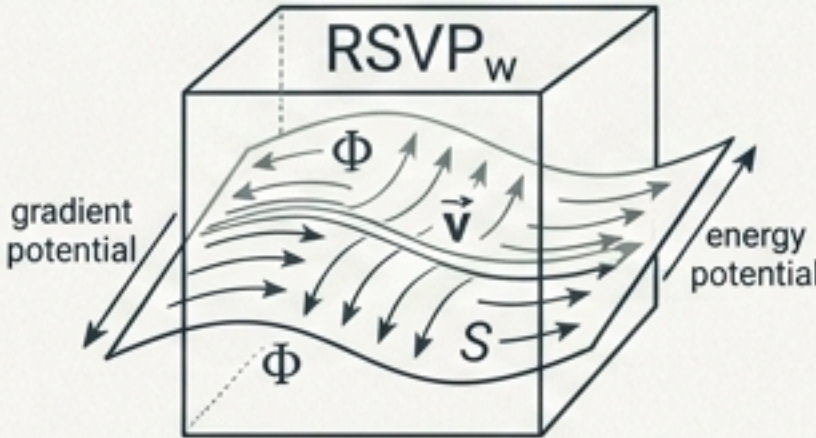
How does structure persist through change?



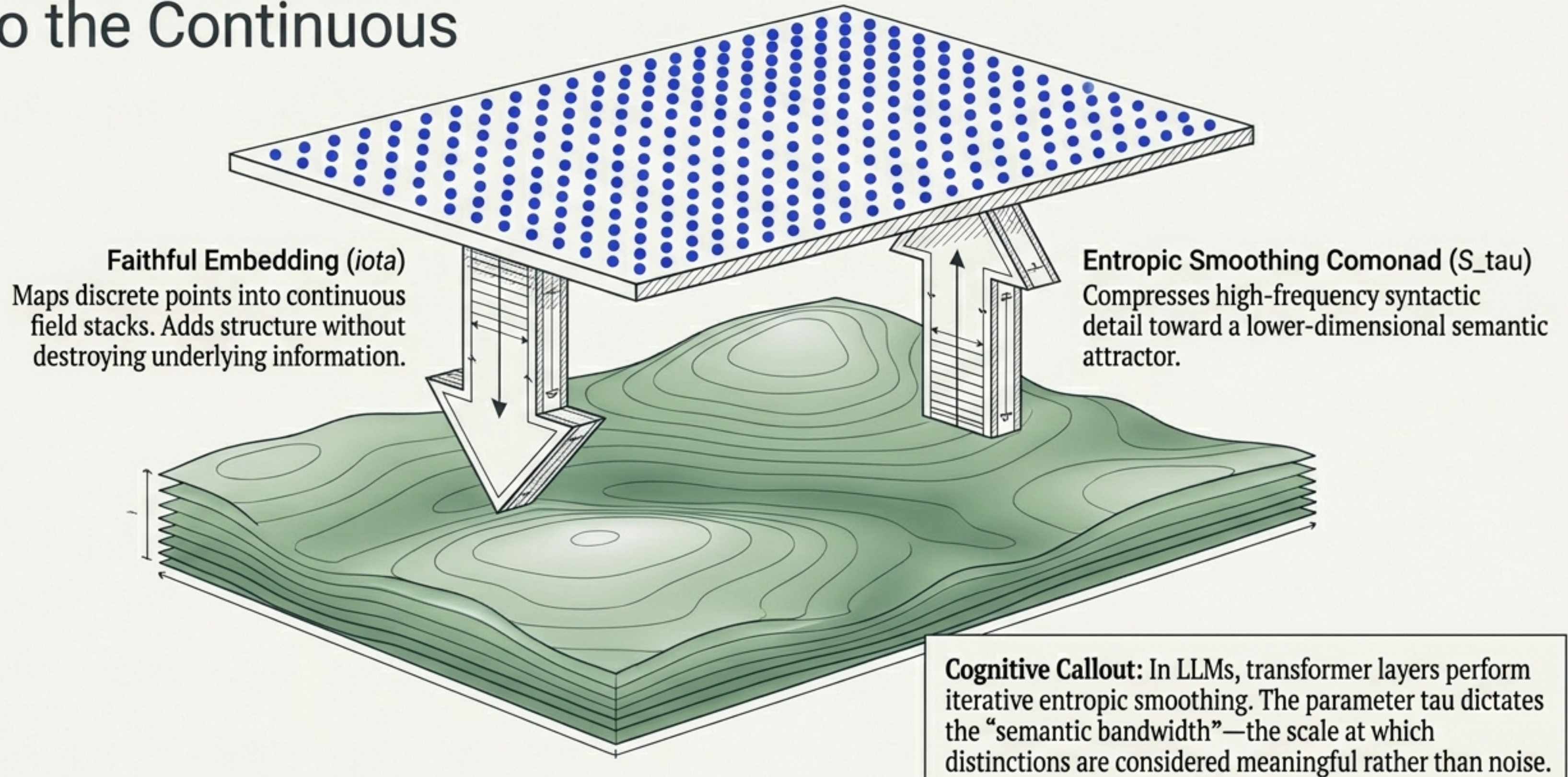
Whether it is an LLM maintaining semantic coherence across 10,000 tokens, a cellular automaton generating persistent life-forms, or a physical field conserving momentum—the mechanism of persistence has historically required separate languages (computability vs. calculus).

The Thesis: Global cognitive and computational structure emerges from local constraint satisfaction. They are mathematically identical.

The Three Pillars of Unification

Discrete Autoregressive Systems (AR _{fin})	Continuous Field Dynamics (RSVP)	Constraint-First Ontology
		
• Focus: LLMs, Cellular Automata. • State at $t+1$ depends on context k. • Key property: Entropy is non-increasing.	• Focus: Relativistic Scalar-Vector Plenum. • Operates on scalar density (Φ), vector flow (v), and entropy (S). • Evolves on a continuous manifold.	• Focus: The new paradigm. • Shifts focus away from what state a system is in. • Focuses on what states are reachable and coherent given an epistemic context.

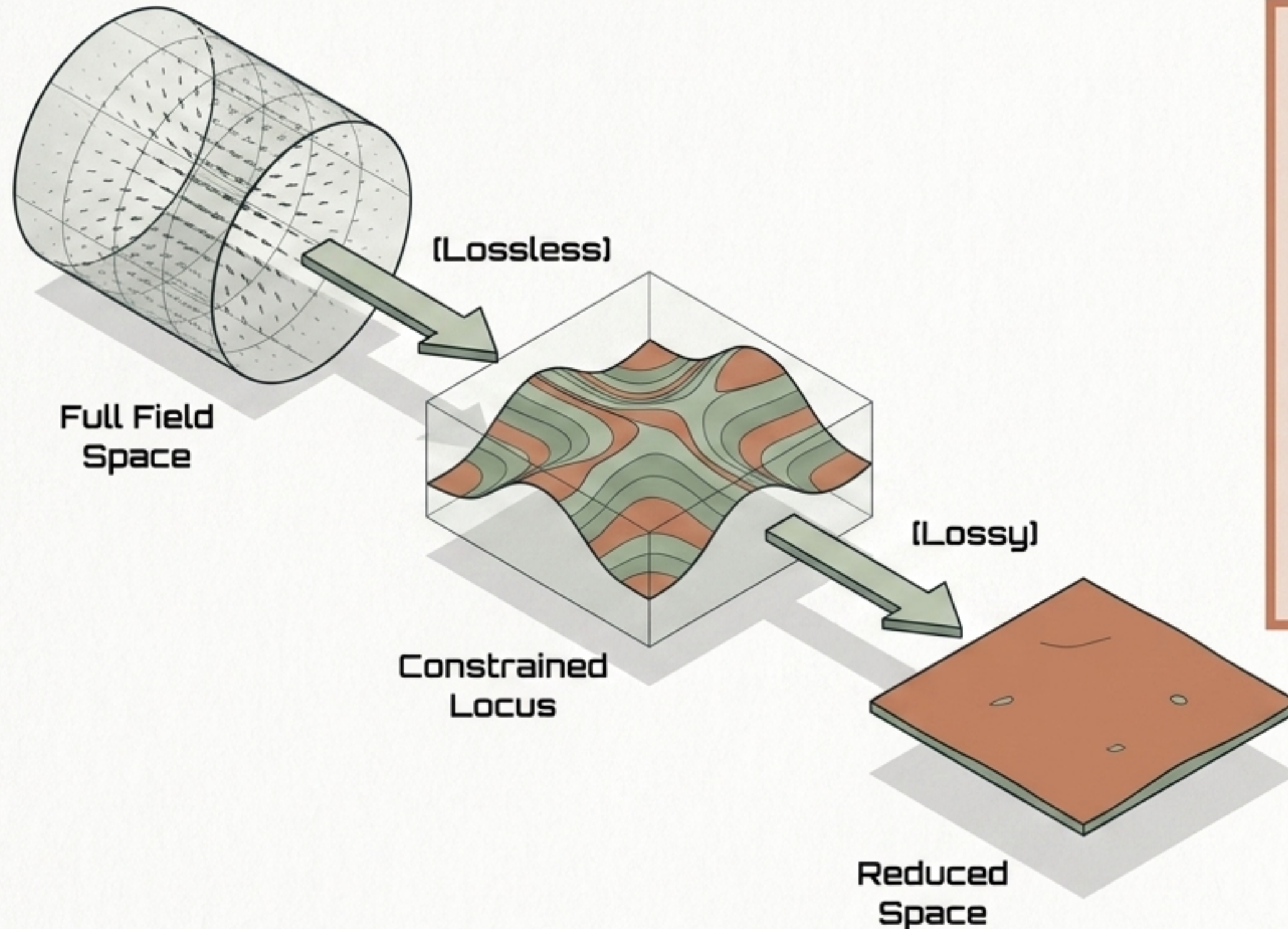
Translating the Discrete into the Continuous



Filtering Semantic Noise

The Process:

The system intentionally discards sub-tau scale fluctuations (gauge directions).



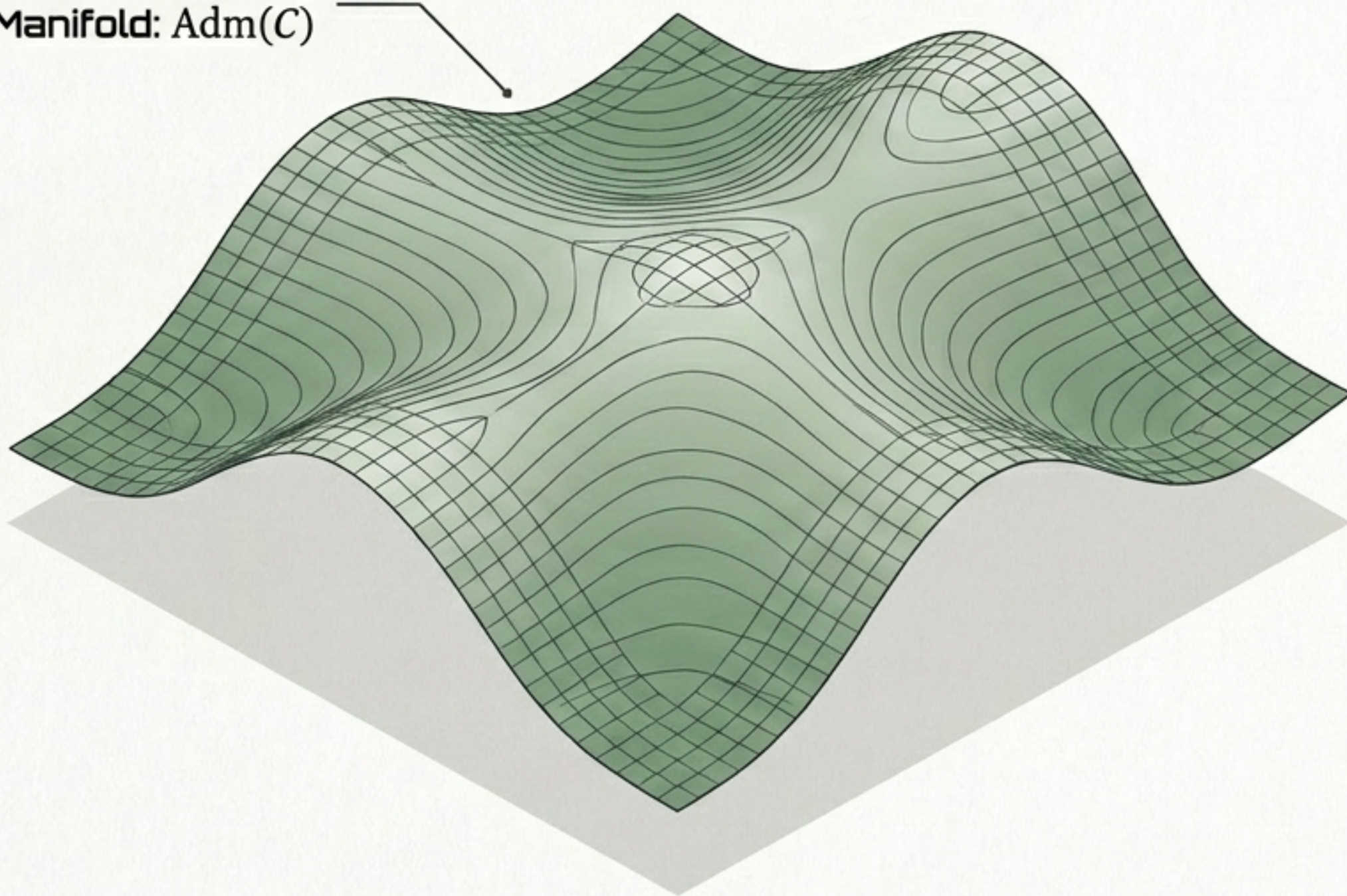
Insight:

The information discarded is the 'reduction entropy.'

Operating at scale tau means intentionally sacrificing high-resolution noise to capture and stabilize pure semantic meaning.

The Space of Coherent Realities

The Admissibility
Manifold: $\text{Adm}(C)$



Core Concept:

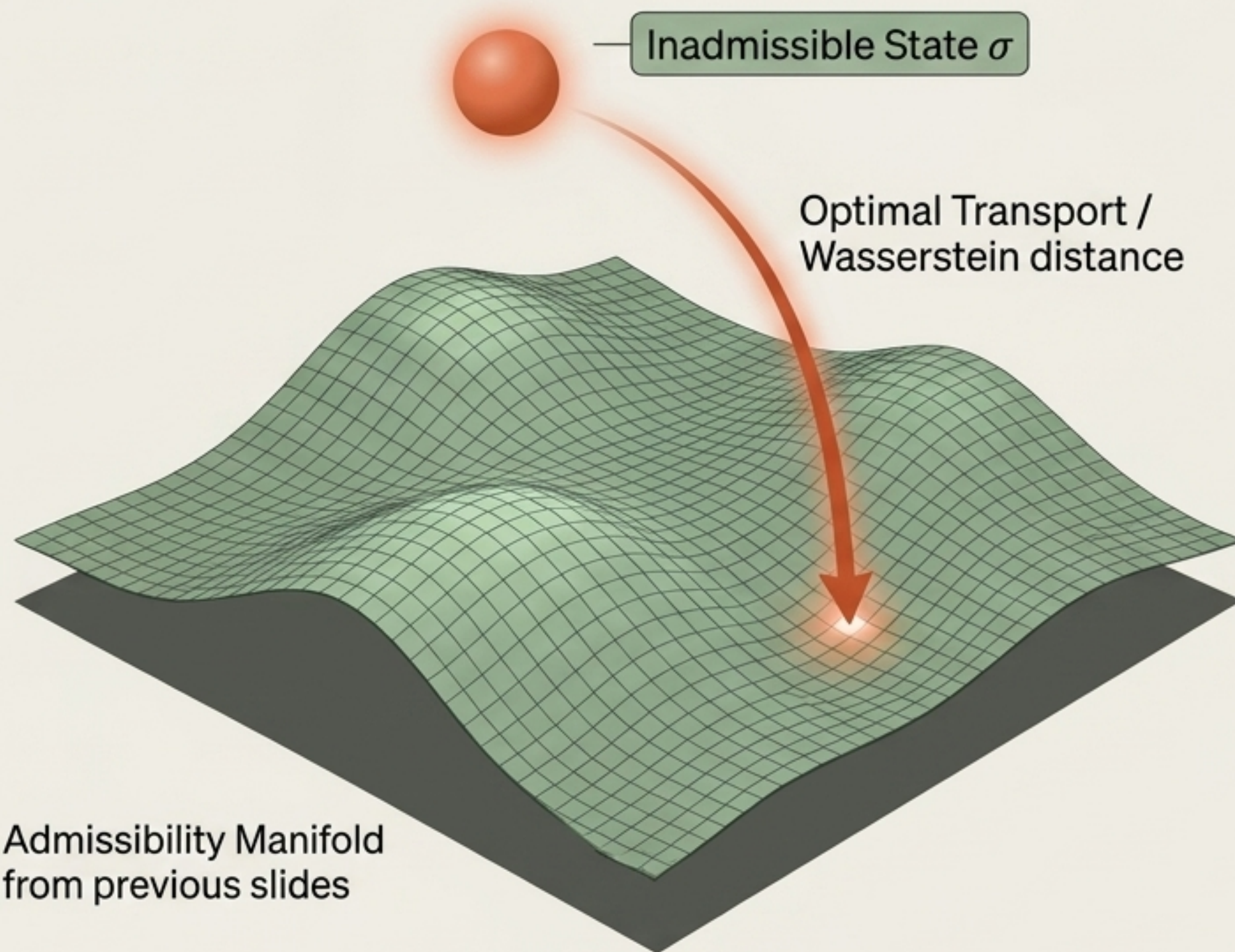
The fundamental object of study is not the state of the system, but the constraints defining what is possible.

Definition:

The sub-manifold of field configurations consistent with a given epistemic context.

- **Physics:** The energy surface of Hamiltonian mechanics.
- **Logic:** The model space of a formal theory.
- **AI:** The valid text distribution in LLM training data.

The Cost of Restoring Coherence



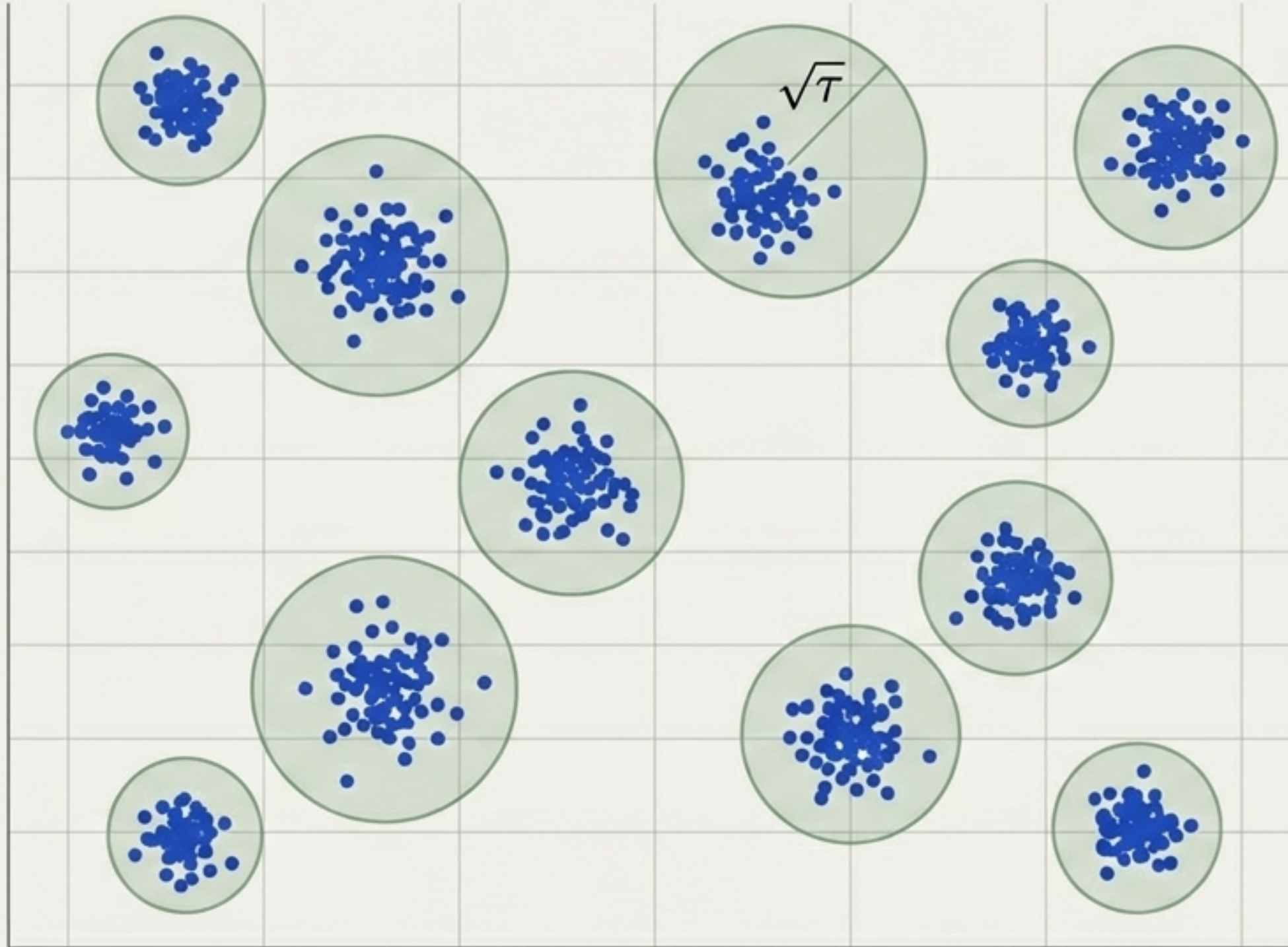
Concept: Perturbations (noise, adversarial inputs, semantic drift) push a cognitive system off its Admissibility Manifold.

The Metric:

The Repair Pseudometric (d_{rep}) is the W_2 Wasserstein distance required to 'repair' the state back to admissibility.

Takeaway: Distances in meaning-space are measured by the energy required to fix them.

The Limits of Resolution



Concept:

Two states are only distinguishable if their repair distance is greater than zero at scale τ . Points inside the same circle are semantically identical.

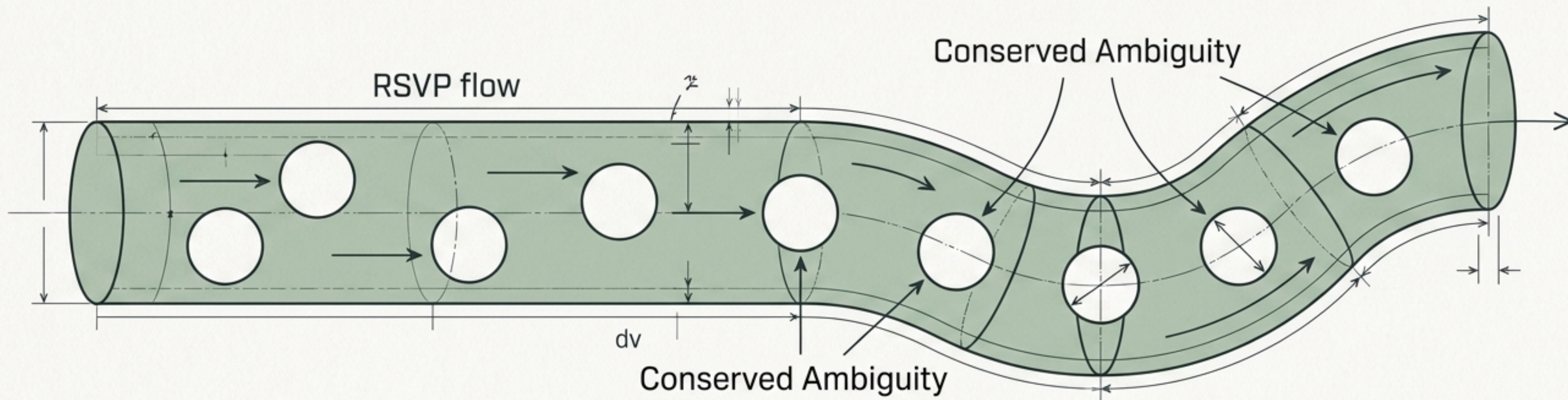
The Math:

Distinction space has finite Minkowski dimension bounded by n/τ .

Cognitive Insight

A mind operating at semantic scale τ can maintain a bounded number of concepts. Increasing nuance (shrinking the circles) costs exponential computational resources.

What Cannot Be Resolved



Observability as Restorability

We do not access states directly; we only observe if a state can be mathematically repaired from given evidence.

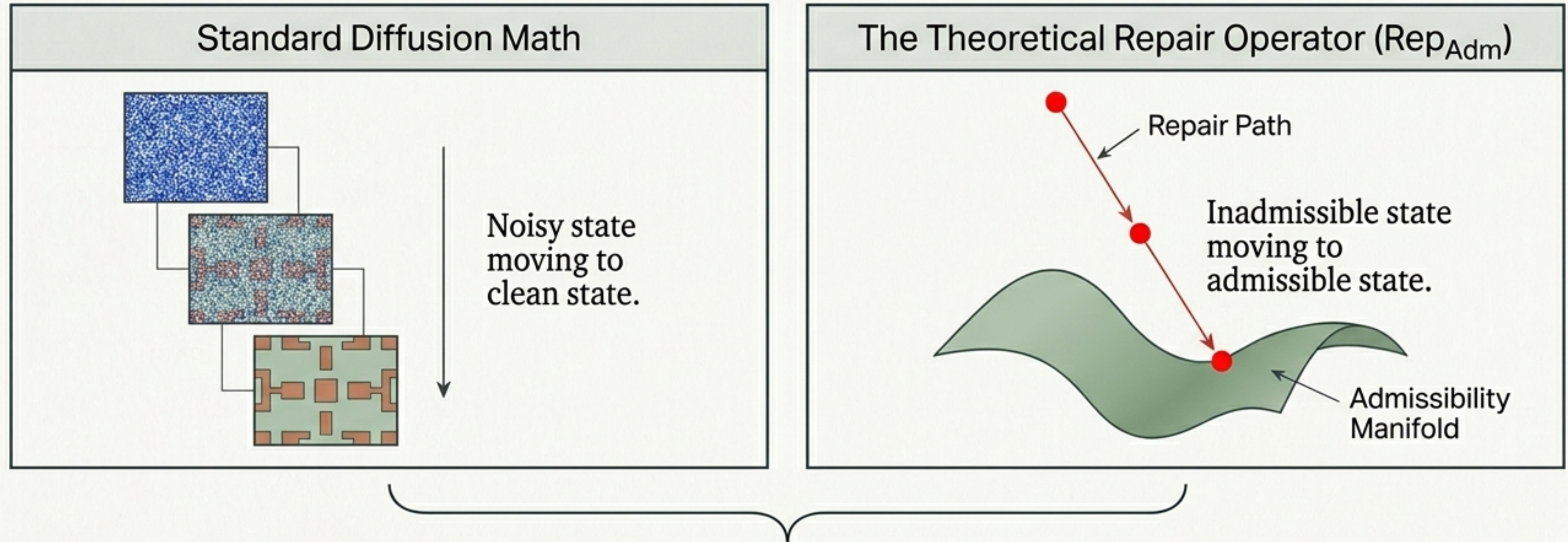
Irreducible Ambiguity

The latent uncertainty that cannot be resolved by any observation within the current epistemic context.

Theorem

Under admissible transformations, irreducible ambiguity is conserved—it acts as an unbreakable topological invariant of the system.

Diffusion Models are Pure Repair Operators



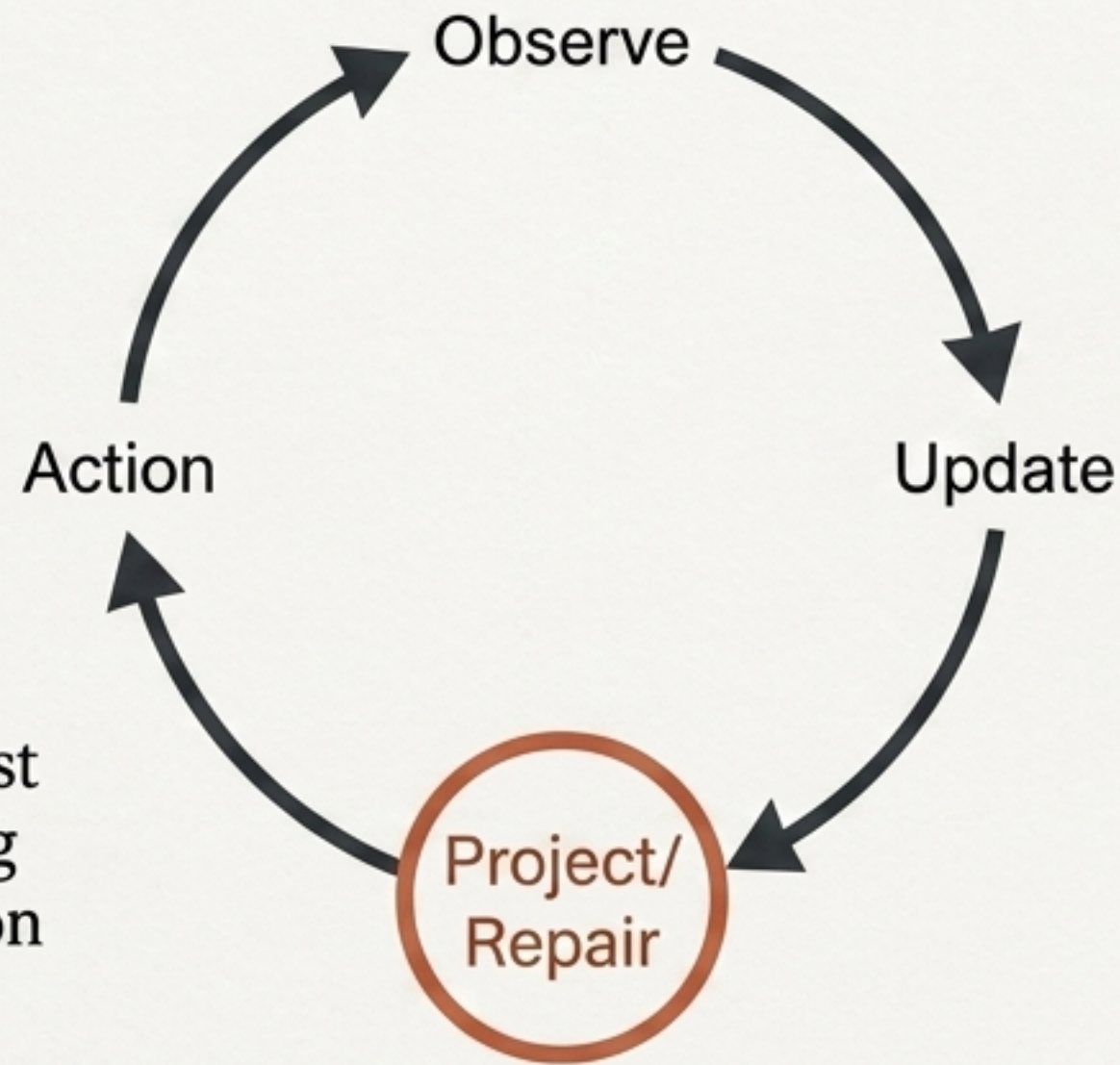
The Link: The reverse-diffusion process in architectures like PHYSIFORMER mathematically mirrors the **exact optimal transport** process of the repair metric d_{rep} .

Implication: Training a diffusion model on physical data inherently learns the Admissibility Manifold. A high repair cost instantly flags an out-of-distribution anomaly.

Agency as Admissibility Restoration

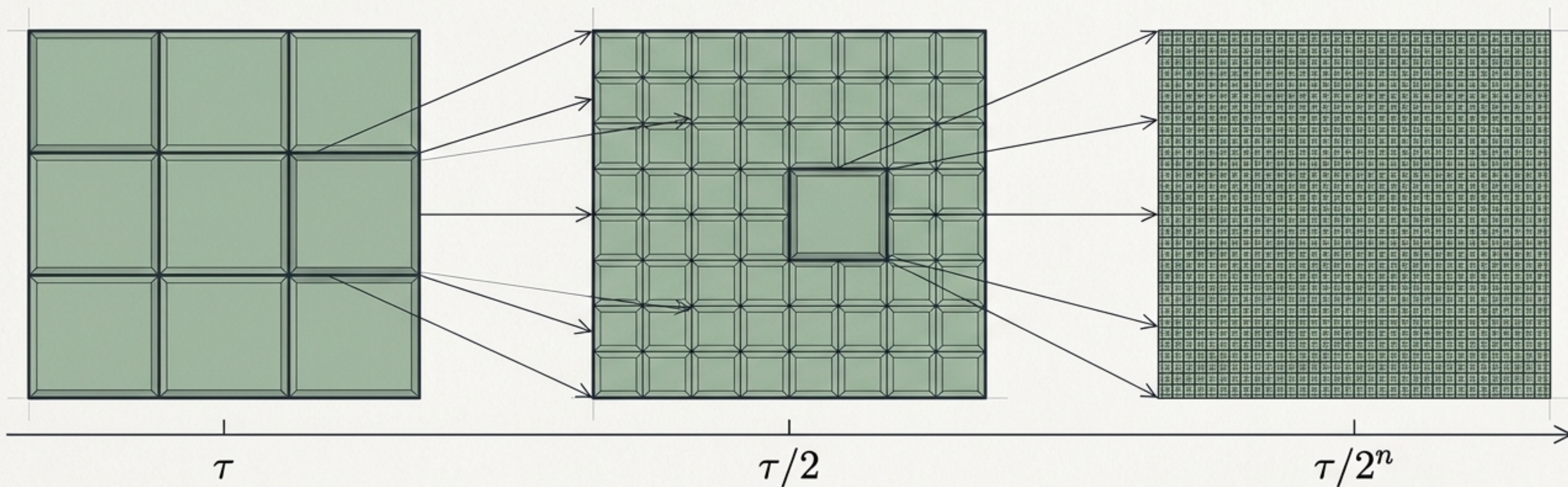
Concept: Consciousness and simulated agency are modeled as a closed loop attempting to minimize “surprise” (Variational Free Energy).

The Resolution: The system must execute massive repair (updating internal beliefs) or physical action (changing the outside world) to restore stability.



The Translation: Surprise is exactly the Repair Metric (d_{rep}). A highly surprising observation pushes the system far off the manifold.

Variable Semantic Granularity



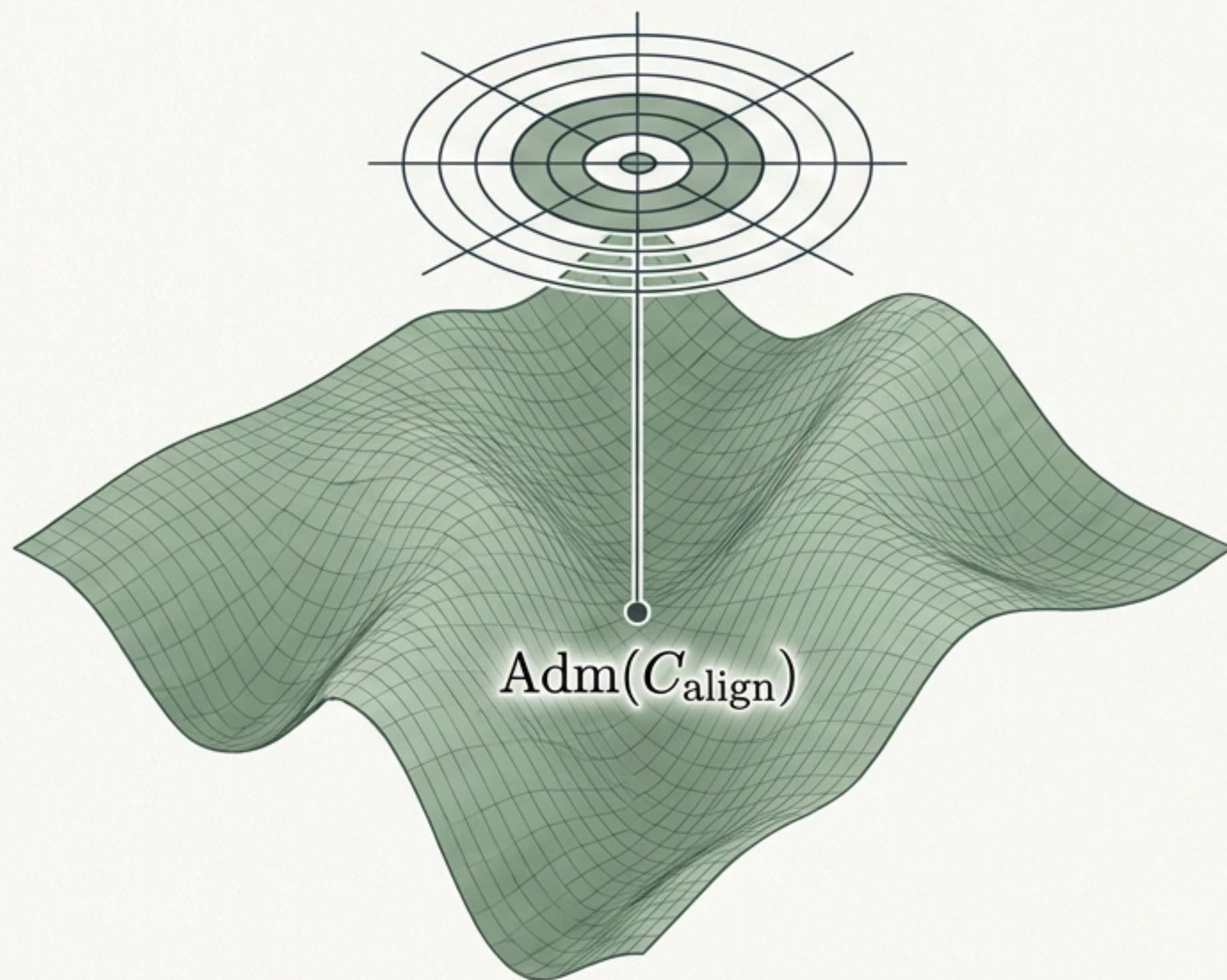
TARTAN Architecture

(Trajectory-Aware Recursive Tiling). Adjusts cognitive granularity by iteratively refining the manifold at finer mathematical scales.

Chain of Memory (CoM)

Links memory states across time. Tracking positions on the admissibility manifold reveals that a massive repair cost between M_t and M_{t+1} signals a memory discontinuity or 'amnesia event'.

Alignment is Restorability



The Framework:

Define target behavior by encoding desired human values explicitly as an admissibility manifold $\text{Adm}(C_{\text{align}})$.

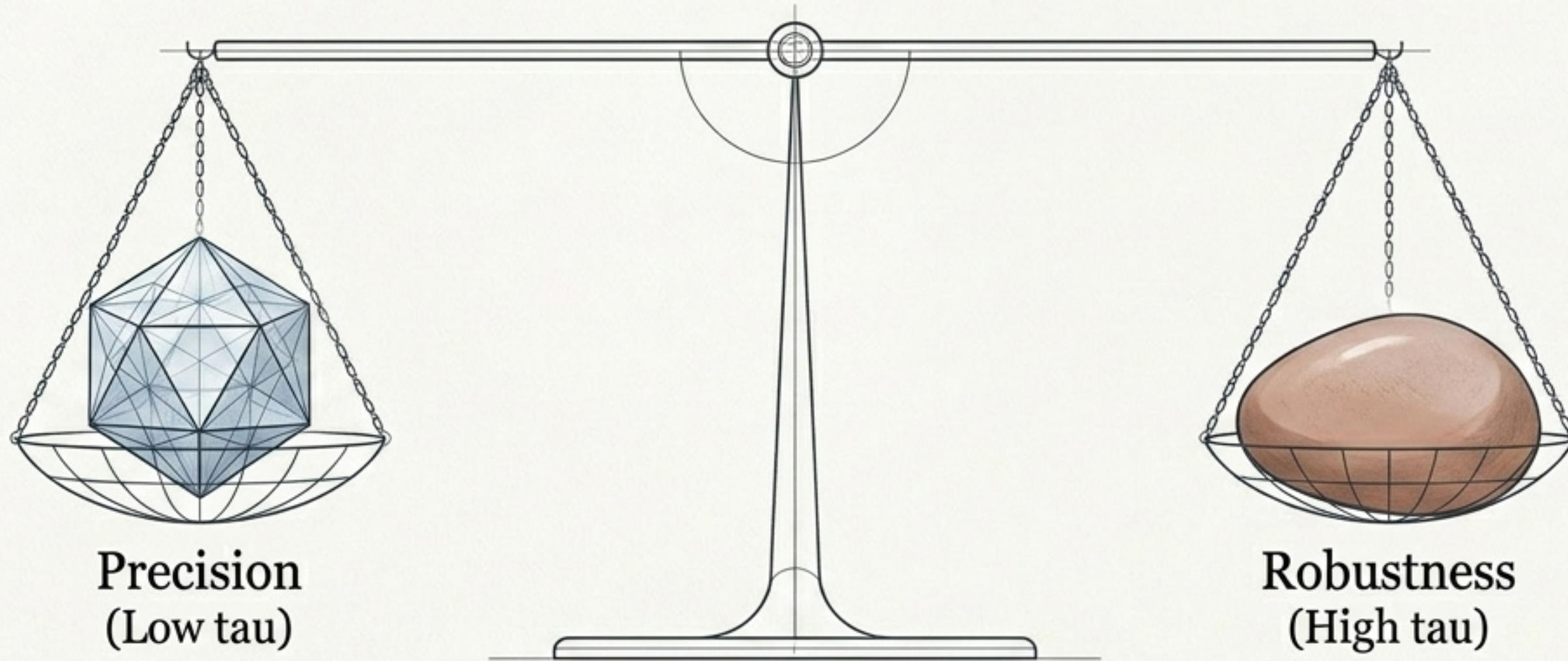
The Method:

Train models to strictly minimize the repair cost d_{rep} as an auxiliary objective during generation.

The Result:

An aligned system is simply one that can always be restored to admissible behavior at a remarkably low computational cost.

The Ecology of the Mind



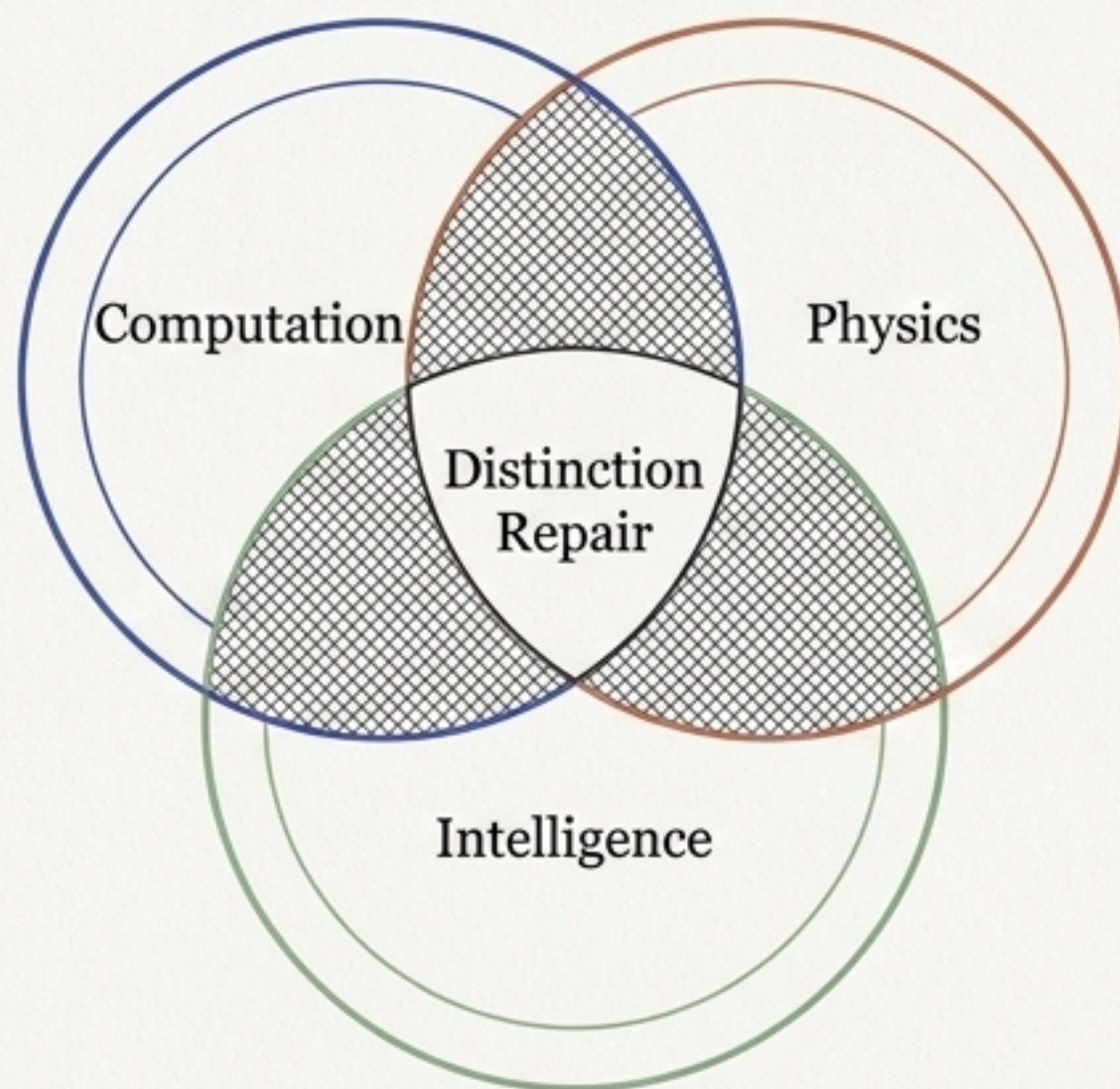
Consciousness

Defined functionally as stable membership in a contextually rich admissibility manifold.
Dissociation and psychosis are mathematically equivalent to sustained high repair costs.

Distinction Ecology

Distinctions compete for finite cognitive resources. Systems must constantly balance fine-grained distinctions (computationally costly, fragile) against coarse-grained ones (cheap, highly robust).

The Ultimate Unification



Summary: Cognition, computation, and physical dynamics are all instances of a single foundational process:
The maintenance and restoration of structured distinctions within a constrained field.

Final Insight: Intelligence is not the ability to represent the world accurately. Intelligence is the ability to repair
the world's representation efficiently, persistently, and at the minimal cost of distinction.